50Days-coding-challenge

Day-20[SQL]

Medical Insurance Cost Analysis

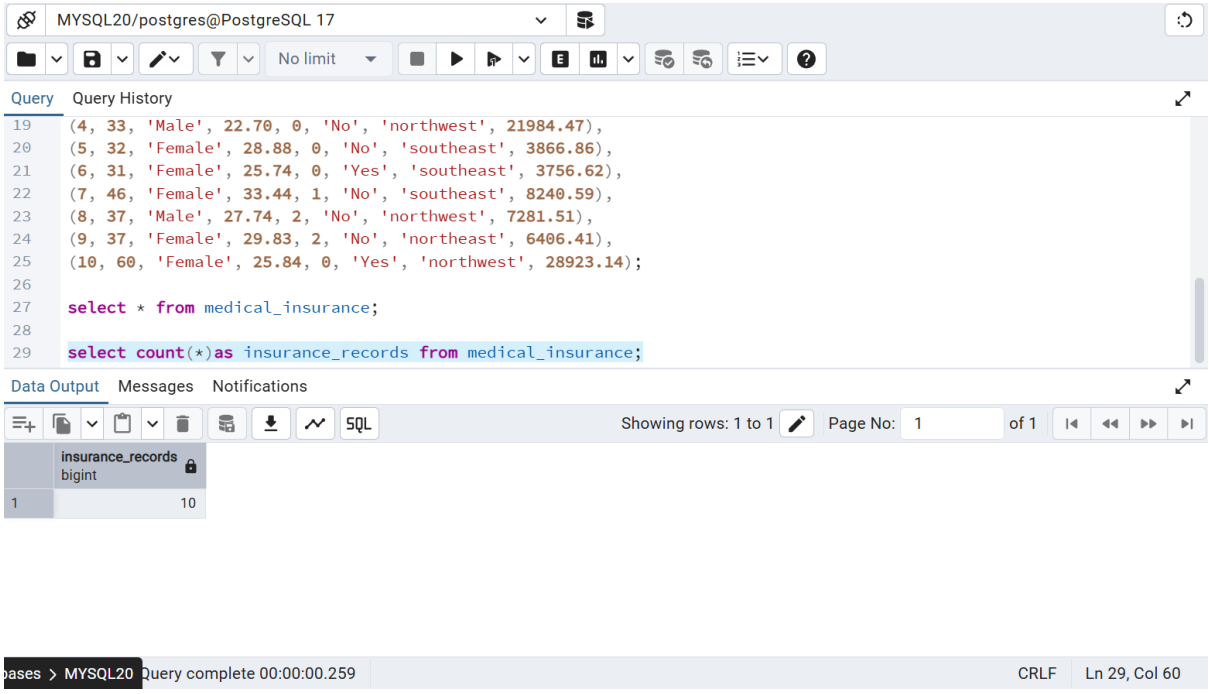
Name:M NITHIN

Create a table in MySQL:

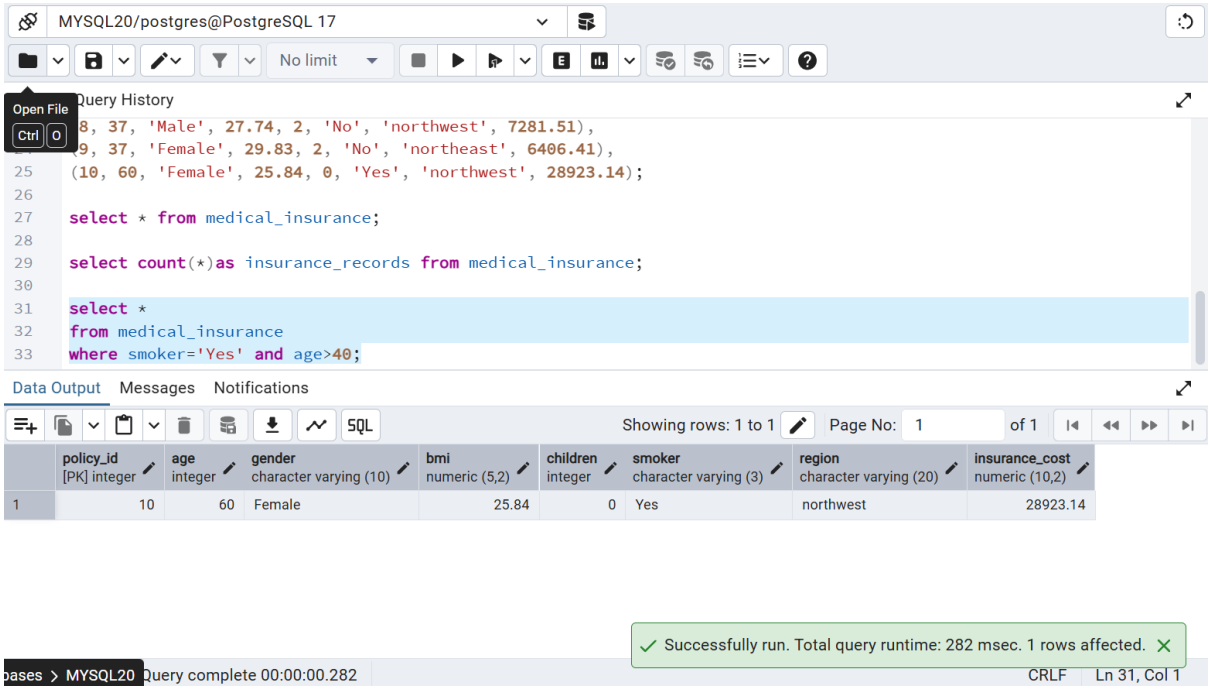
```
CREATE TABLE medical_insurance (     policy_id INT PRIMARY KEY,
age INT,     gender VARCHAR(10),     bmi DECIMAL(5,2),     children
INT,     smoker VARCHAR(3),       -- Yes / No     region
VARCHAR(20),       -- e.g., northeast, northwest, southeast,
southwest     insurance_cost DECIMAL(10,2)
);
INSERT INTO medical_insurance VALUES
(1, 19, 'Female', 27.90, 0, 'Yes', 'southwest', 16884.92),
(2, 18, 'Male', 33.77, 1, 'No', 'southeast', 1725.55),
(3, 28, 'Male', 33.00, 3, 'No', 'southeast', 4449.46),
(4, 33, 'Male', 22.70, 0, 'No', 'northwest', 21984.47),
(5, 32, 'Female', 28.88, 0, 'No', 'southeast', 3866.86),
(6, 31, 'Female', 25.74, 0, 'Yes', 'southeast', 3756.62),
(7, 46, 'Female', 33.44, 1, 'No', 'southeast', 8240.59),
(8, 37, 'Male', 27.74, 2, 'No', 'northwest', 7281.51),
(9, 37, 'Female', 29.83, 2, 'No', 'northeast', 6406.41),
(10, 60, 'Female', 25.84, 0, 'Yes', 'northwest',
28923.14);
```

Tasks

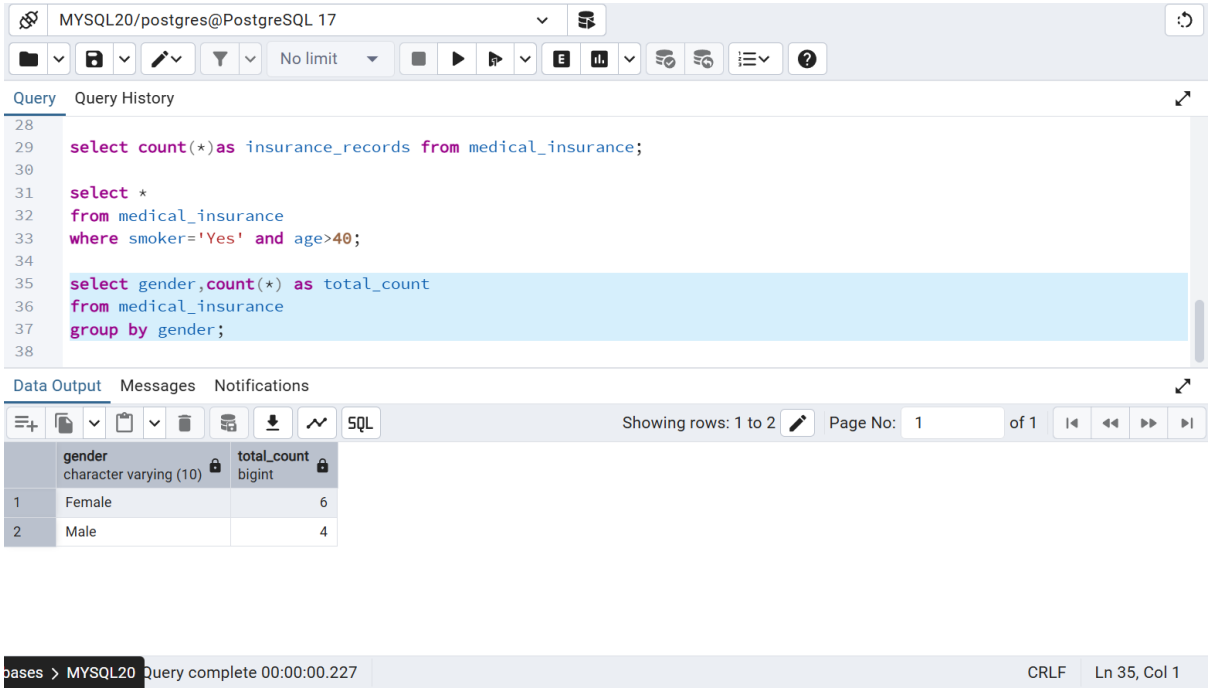
1. Find the **total number of insurance records**.



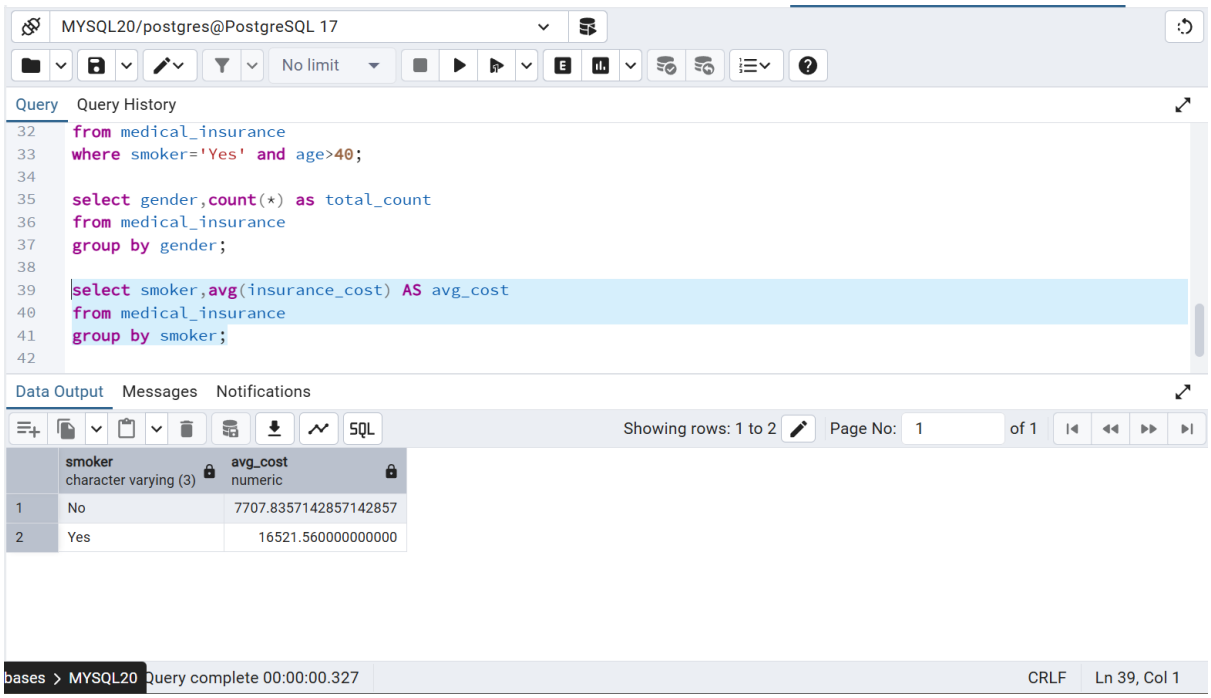
2. Retrieve the details of all **smokers above age 40**.



3. Count how many **male and female policyholders** exist in the dataset.



4. Calculate the **average insurance cost** of smokers vs. non-smokers.



5. Find the **region with the highest average insurance cost**.

MYSQL20/postgres@PostgreSQL 17

Query | Query History

```sql
group by gender;

select smoker,avg(insurance_cost) AS avg_cost
from medical_insurance
group by smoker;

SELECT region, AVG(insurance_cost) AS avg_cost
FROM medical_insurance
GROUP BY region
ORDER BY avg_cost DESC
LIMIT 1;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	region character varying (20)	avg_cost numeric
1	northwest	19396.373333333333

Total rows: 1 Query complete 00:00:00.448 CRLF Ln 43, Col 1

6. List the **top 5 policyholders who paid the highest insurance cost**.

MYSQL20/postgres@PostgreSQL 17

Query | Query History

```sql
SELECT region, AVG(insurance_cost) AS avg_cost
FROM medical_insurance
GROUP BY region
ORDER BY avg_cost DESC
LIMIT 1;

SELECT *
FROM medical_insurance
ORDER BY insurance_cost DESC
LIMIT 5;
```

Data Output | Messages | Notifications

Showing rows: 1 to 5 Page No: 1 of 1

	policy_id [PK] integer	age integer	gender character varying (10)	bmi numeric (5,2)	children integer	smoker character varying (3)	region character varying (20)	insurance_cost numeric (10,2)
1	10	60	Female	25.84	0	Yes	northwest	28923.14
2	4	33	Male	22.70	0	No	northwest	21984.47
3	1	19	Female	27.90	0	Yes	southwest	16884.92
4	7	46	Female	33.44	1	No	southeast	8240.59
5	8	37	Male	27.74	2	No	northwest	7281.51

Total rows: 5 Query complete 00:00:00.264 CRLF Ln 49, Col 1

7. Calculate the **average insurance cost grouped by age group**:

   Young (18–30), Middle-aged (31–50), Senior (51+).

MYSQL20/postgres@PostgreSQL 17

Query | Query History

```sql
LIMIT 5;

SELECT
    CASE
        WHEN age BETWEEN 18 AND 30 THEN 'Young'
        WHEN age BETWEEN 31 AND 50 THEN 'Middle-aged'
        ELSE 'Senior'
    END AS age_group,
    AVG(insurance_cost) AS avg_cost
FROM medical_insurance
GROUP BY age_group;
```

Data Output | Messages | Notifications

Showing rows: 1 to 3 | Page No: 1 of 1

	age_group text	avg_cost numeric
1	Middle-aged	8589.4100000000000000
2	Young	7686.6433333333333333
3	Senior	28923.140000000000

Total rows: 3 | Query complete 00:00:00.423 | CRLF | Ln 54, Col 1

8. Identify if **smokers with BMI > 30** are paying significantly higher insurance than nonsmokers.

MYSQL20/postgres@PostgreSQL 17

Open File Ctrl O

Query History

```sql
        WHEN age BETWEEN 31 AND 50 THEN 'Middle-aged'
        ELSE 'Senior'
    END AS age_group,
    AVG(insurance_cost) AS avg_cost
FROM medical_insurance
GROUP BY age_group;

SELECT smoker, AVG(insurance_cost) AS avg_cost
FROM medical_insurance
WHERE bmi > 30
GROUP BY smoker;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 | Page No: 1 of 1

	smoker character varying (3)	avg_cost numeric
1	No	4805.2000000000000000

Total rows: 1 | Query complete 00:00:00.164 | CRLF | Ln 66, Col 15

9. Find the **correlation-like insights** (not actual correlation in SQL, but using GROUP BY) – e.g.,

- Avg cost by number of children.

MYSQL20/postgres@PostgreSQL 17

Query | Query History

```sql
GROUP BY age_group;

SELECT smoker, AVG(insurance_cost) AS avg_cost
FROM medical_insurance
WHERE bmi > 30
GROUP BY smoker;

SELECT children, AVG(insurance_cost) AS avg_cost
FROM medical_insurance
GROUP BY children
ORDER BY children;
```

Data Output | Messages | Notifications

Showing rows: 1 to 4 Page No: 1 of 1

	children integer	avg_cost numeric
1	0	15083.202000000000
2	1	4983.0700000000000000
3	2	6843.9600000000000000
4	3	4449.4600000000000000

Total rows: 4 | Query complete 00:00:00.199 | CRLF | Ln 69, Col 1

- Avg cost by gender.

MYSQL20/postgres@PostgreSQL 17

Query | Query History

```sql
WHERE bmi > 30
GROUP BY smoker;

SELECT children, AVG(insurance_cost) AS avg_cost
FROM medical_insurance
GROUP BY children
ORDER BY children;

SELECT gender, AVG(insurance_cost) AS avg_cost
FROM medical_insurance
GROUP BY gender;
```

Data Output | Messages | Notifications

Showing rows: 1 to 2 Page No: 1 of 1

	gender character varying (10)	avg_cost numeric
1	Female	11346.4233333333333333
2	Male	8860.2475000000000000

Total rows: 2 | Query complete 00:00:00.178 | CRLF | Ln 74, Col 1

- Avg cost by region.

MYSQL20/postgres@PostgreSQL 17

Query | Query History

```sql
FROM medical_insurance
GROUP BY children
ORDER BY children;

SELECT gender, AVG(insurance_cost) AS avg_cost
FROM medical_insurance
GROUP BY gender;

SELECT region, AVG(insurance_cost) AS avg_cost
FROM medical_insurance
GROUP BY region;
```

Data Output | Messages | Notifications

Showing rows: 1 to 4 Page No: 1 of 1

	region character varying (20)	avg_cost numeric
1	northwest	19396.373333333333
2	southeast	4407.8160000000000000
3	northeast	6406.4100000000000000
4	southwest	16884.9200000000000000

Total rows: 4 Query complete 00:00:00.179 CRLF Ln 78, Col 1

10. Write a query to find the **insurance cost difference between smokers and non-smokers in each region**.

MYSQL20/postgres@PostgreSQL 17

Query | Query History

```sql
FROM medical_insurance
GROUP BY region;

SELECT
    region,
    AVG(CASE WHEN smoker = 'Yes' THEN insurance_cost END) AS smoker_avg,
    AVG(CASE WHEN smoker = 'No' THEN insurance_cost END) AS non_smoker_avg,
    (AVG(CASE WHEN smoker = 'Yes' THEN insurance_cost END) -
     AVG(CASE WHEN smoker = 'No' THEN insurance_cost END)) AS cost_difference
FROM medical_insurance
GROUP BY region;
```

Data Output | Messages | Notifications

Showing rows: 1 to 4 Page No: 1 of 1

	region character varying (20)	smoker_avg numeric	non_smoker_avg numeric	cost_difference numeric
1	northwest	28923.140000000000	14632.9900000000000000	14290.1500000000000000
2	southeast	3756.6200000000000000	4570.6150000000000000	-813.9950000000000000
3	northeast	[null]	6406.4100000000000000	[null]
4	southwest	16884.9200000000000000	[null]	[null]

Total rows: 4 Query complete 00:00:00.330 CRLF Ln 82, Col 1